Pooja Patel

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Profile Summary

Computer Science graduate and aspirant with a robust foundation in core CS fundamentals, distributed architectures, and cybersecurity. Deeply passionate about tackling complex engineering challenges at the intersection of Machine Learning, Systems Design, and Security Infrastructure.

Education

Parul University **Vadodara, India**

BTech in Computer Science Engineering, CGPA: 8.0/10 (First Class with Distinction) *2022–2026*

Strong focus on CS fundamentals, Security, and Operating Systems preparation.

Technical Skills

Languages: Python, Ruby (learning), JavaScript, C/C++, Bash

Applied ML & AI: Amazon Bedrock (Nova Pro, Claude 4 Sonnet), TinyLLM, NLP Analysis, Predictive Modeling

Systems & Architecture: FastAPI, Redis Streams (learning) (Event-Driven/Distributed), Neo4j (Knowledge Graphs), AST Analysis

Cloud & DevOps: Docker Compose, Kubernetes, MinIO (learning) (Data Lakes), Google Cloud, CI/CD, GitHub Actions

Security & Core: Vulnerability Assessment, Asset Discovery, Sandbox Execution (in practice), Threat Modelling

Professional Experience

Udaan Society **Remote**

Software Development Engineer *Jan 2026–Mar 2026*

- Engineered a custom, scalable project management platform tailored for non-profit operations, streamlining task allocation, resource tracking, and cross-functional team collaboration. While handling leadership of team of fellow developers.

Applied Research & Systems Projects

Distributed Cybersecurity Platform

NETRA v2 - AI-Native Asset Discovery Engine *2025*

- **Distributed Architecture:** Architected an event-driven reconnaissance pipeline utilising Redis Streams to decouple heavy I/O ingestion from CPU-intensive ML analysis.
- **Knowledge Graph & Storage:** Integrated Neo4j for real-time dynamic asset mapping and visualisation, alongside MinIO for highly scalable, long-term data lake storage.
- **ML Integration:** Developed a 'Zombie API' detection mechanism using a hybrid Ruby/Python engine and TinyLLM for shadow API discovery, paired with an NLP-powered 'Cyber-AI Neural Terminal'.

AI-for-Bharat Hackathon

OASIS - Self-Healing Documentation Engine *2026*

- **Automated Validation System:** Developed an automated Git bot that utilises static AST analysis and isolated runtime sandboxing to validate code snippets across 10+ programming languages.
- **AI-Powered Auto-Correction:** Implemented an auto-correction pipeline utilising Amazon Bedrock (Nova Pro + Claude 4 Sonnet) to detect breaking changes and dynamically heal repository documentation.
- **Workflow Integration:** Engineered seamless CI/CD integration into GitHub/GitLab PR workflows to maintain a mathematically accurate Single Source of Truth for technical documentation.

Final year Project

Chronovault *2025*

- Engineered Chronovault, a hybrid full-stack web application powered by Google SDKs, Firebase, and map APIs that bridges the gap between digital self-guided care and traditional clinical therapy; the platform combines evidence-based features—including an AI-driven 'TimeCapsule' sentiment journaling engine, calming audio streaming, and wellness workshop listings—with an administrative database and intuitive dashboard that allows partner doctors to efficiently manage localized patient appointments.

Leadership & Volunteering

National Cadet Corps (NCC)

Cadet *2022–2025*

Led civic campaigns on hygiene and environmental awareness. Developed robust crisis management, public engagement, and technical leadership skills in high-pressure environments.